

MID-STATE TECHNICAL COLLEGE • IT PROGRAMS

Connecting the Firewall with Security Zones

Unit 5 • Information Security 2 (10-151-111)



Learning Outcomes

By the end of this unit, you will be able to:



Describe network segmentation components

Explain how zones, interfaces, virtual routers, and policy combine to block threats.



Configure security zones

Create zones and assign interfaces to implement network segmentation.



Deploy tap and virtual wire interfaces

Monitor passively from a SPAN port, or insert the firewall inline without readdressing.



Configure Layer 3 interfaces

Assign IP addressing and build VLAN subinterfaces that route between networks.



Configure a virtual router

Attach interfaces, add a static default route, and read the route and forwarding tables.



Apply Interface Management Profiles

Control which management services each data-plane interface answers.

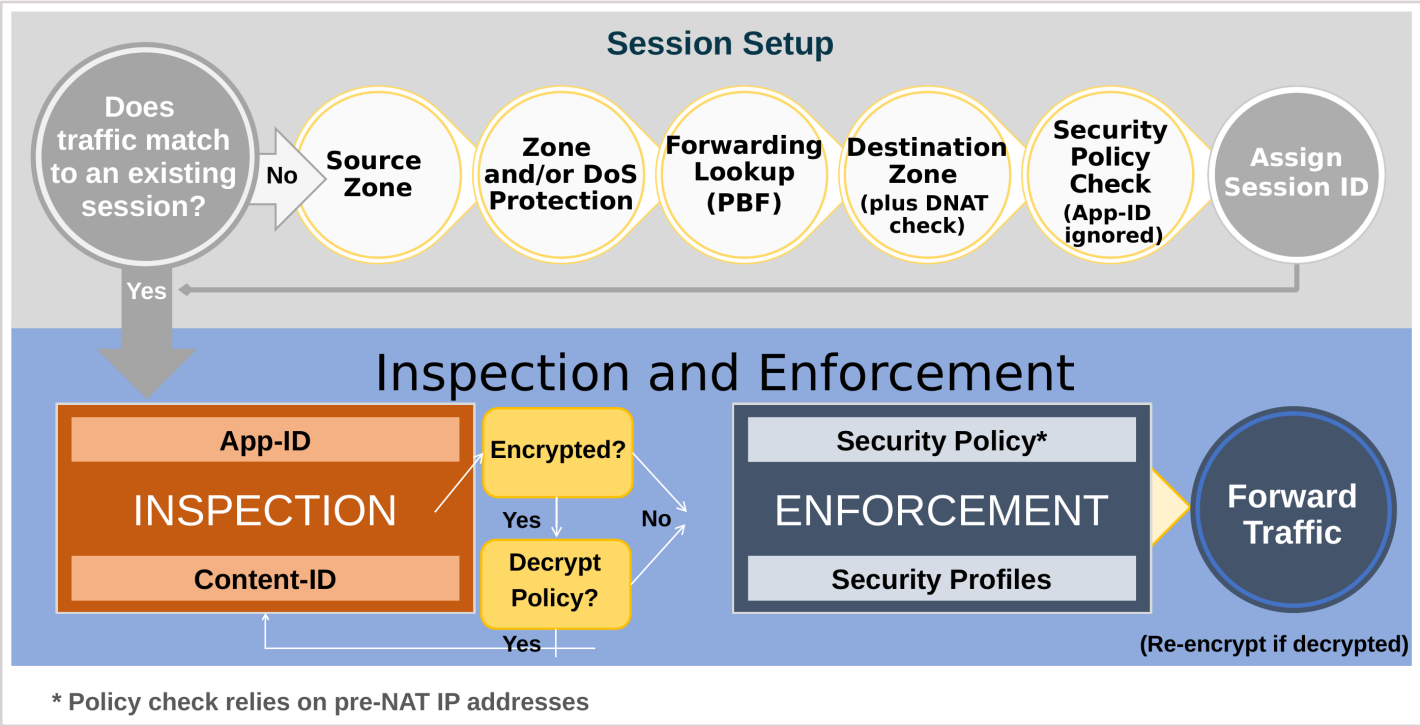
LESSON 1

Segmentation and the Packet Flow



Flow Logic of the Next-Generation Firewall

Every packet that is not part of an existing session goes through the same four stages.



Where zones are decided

Source zone comes from the ingress interface.

Destination zone comes from the route lookup that picks the egress interface.

No zone on the interface means no session — and no traffic.

Network Segmentation

- Divide the network into areas, each protected by the firewall, so compromising one area does not hand over the rest.
- Start with the business: **who must access what** — which groups use which applications and data.
- Apply the **principle of least privilege**: a vendor may need one application, not the whole data center.
- Account for **regulatory requirements** such as PCI DSS that dictate where sensitive data lives and who may reach it.



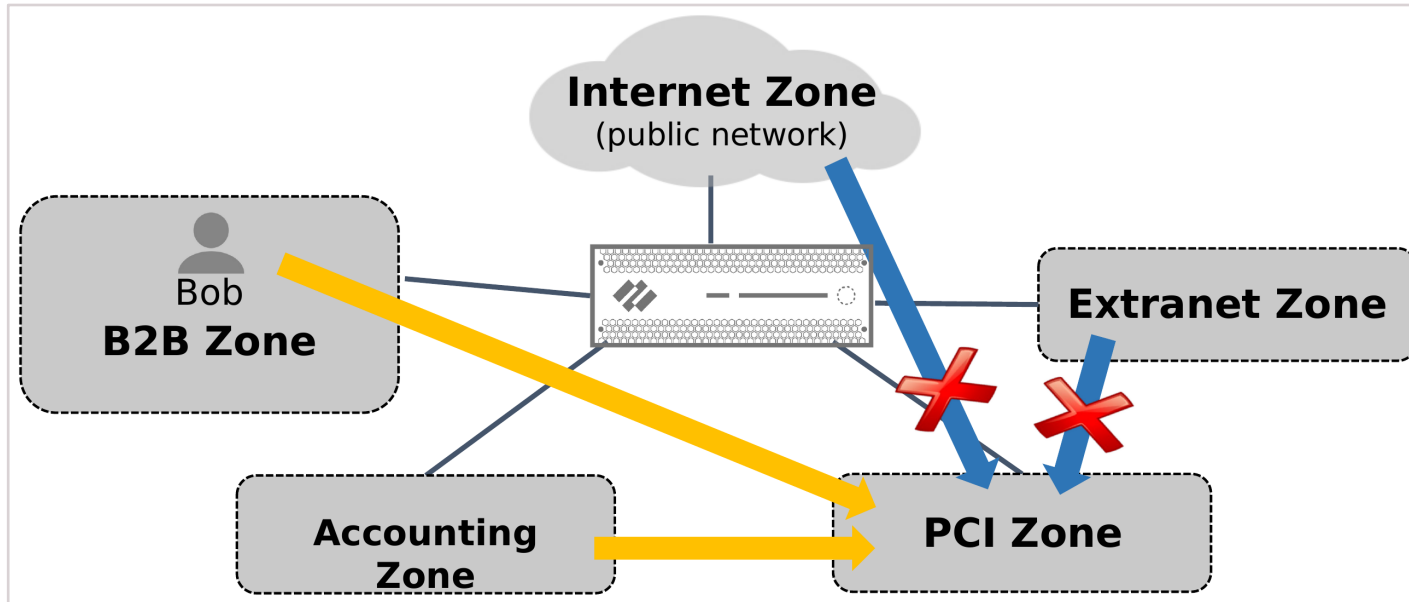
Put crown jewels deepest

Credit card data, employee records, and customer contact data each deserve their own segment.

Business-critical data belongs in the most protected zone, with the fewest paths in.

Security Zones Reduce the Attack Surface

A zone groups devices and users with similar security needs — and usually aligns with a network segment.



- Traffic flows freely **within** a zone.
- Traffic **between** zones is denied until a Security policy rule allows it.
- Smaller, more specific zones mean tighter control and harder lateral movement.

Security Policy Enforces the Segmentation

Policies > Security — two allow rules define every path into the PCI zone.

	NAME	TAGS	TYPE	Source				Destination			APPLICATION	SERVICE	ACTION
				ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRE...	DEVICE			
1	B2B-PCI-Access	B2B	universal	B2B	any	Bob	any	PCI	any	any	mssql-db	application-default	Allow
2	Acct-PCI-Access	Accounting	universal	Accounting	any	Accounting_Grp	any	PCI	any	any	mssql-db	application-default	Allow
3	intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any	any	any	any	Allow
4	interzone-default	none	interzone	any	any	any	any	any	any	any	any	any	Deny

Allowed

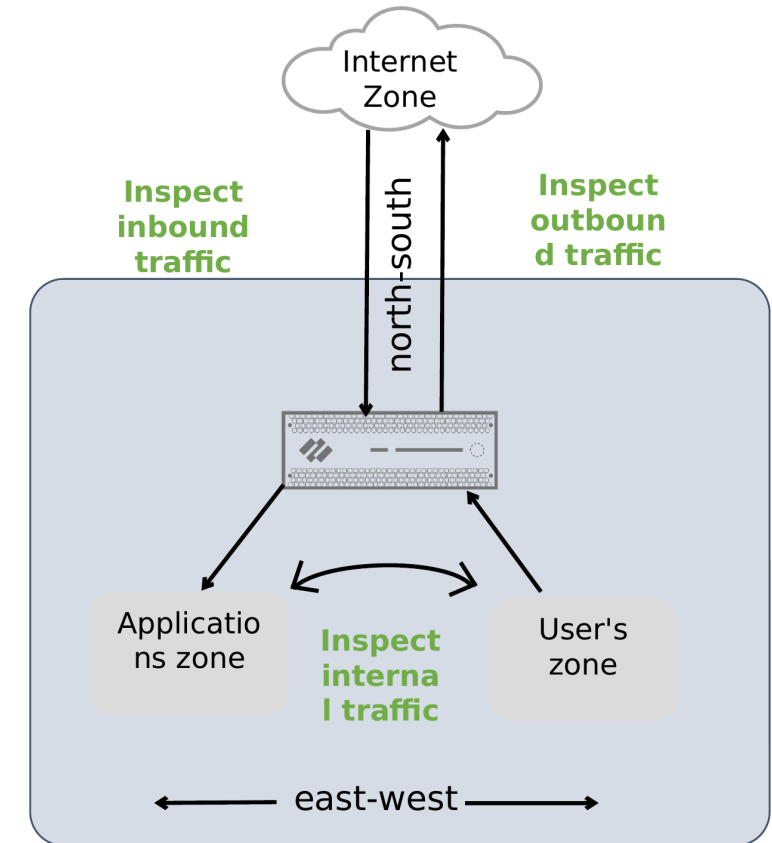
- **Bob** in the B2B zone → PCI zone
- **Accounting-Grp** in the Accounting zone → PCI zone

Blocked by default

- Every other B2B user
- Anyone in the Extranet or Internet zones

Zero Trust Architecture

- **Never trust, always verify** — the inside of the network is not assumed safe.
- Inspect **north-south** traffic at the perimeter: inbound to published services and outbound from users.
- Also inspect **east-west** traffic between internal segments, such as the users' zone and the applications zone.
- Segment with subnets or VLANs, then use zones and policy to control every crossing.





LESSON 2

Interfaces and Security Zones

Network Interfaces

The firewall data plane controls the in-band traffic interfaces, named ethernet slot/port.



Single-slot firewall



Multi-slot firewall



Logical interfaces

- `ethernet1/2` — on a single-slot firewall the first number is always 1; on a chassis it is the slot.
- Each interface belongs to exactly **one zone**; a zone can hold many interfaces of the same type.
- **Subinterfaces** such as `ethernet1/1.1` split one port into logical interfaces, often one per VLAN.

Interface Types and Zone Types

Different zone types accept only specific interface types — the dropdowns enforce the match.

Zone type	Interfaces it accepts	Typical use
Tap	Tap interfaces	Passive visibility from a SPAN or mirror port
Virtual Wire	Virtual wire interfaces	Transparent inline deployment
Layer 2	Layer 2 interfaces	Switching within a broadcast domain
Layer 3	Layer 3, VLAN, loopback, and tunnel interfaces	Routing between networks — this lab
Tunnel	None assigned	Tunnel-in-tunnel content inspection

Not zoned: MGT and HA interfaces are never assigned to a zone — they do not carry production traffic.

Create a Security Zone

Network > Zones > Add — name the zone, choose its type, and assign interfaces.

The screenshot shows the 'Zone' configuration window. At the top, the 'Name' field is set to 'DMZ'. Below it, 'Log Setting' is set to 'None'. The 'Type' dropdown is set to 'Virtual Wire'. Under the 'INTERFACES' section, 'ethernet1/4' and 'ethernet1/5' are selected. The 'Zone Protection' section at the bottom left has 'Zone Protection Profile' set to 'None', with 'Enable Packet Buffer Protection' checked and 'Enable Header Inspection' unchecked. The 'User Identification ACL' section on the right has 'Enable User Identification' unchecked. It contains an 'INCLUDE LIST' with a text input field and 'Add'/'Delete' buttons, and an 'EXCLUDE LIST' with a similar input field and buttons. The 'Device-ID ACL' section on the far right has 'Enable Device Identification' unchecked and similar 'INCLUDE' and 'EXCLUDE' list sections. A help icon (?) is in the top right corner.

- **Name** is case-sensitive — pick a naming standard.
- **Type** must match the interfaces you add.
- Interfaces not in a zone **do not process traffic**.
- Zones can be created from the interface dialog with **New Zone**.

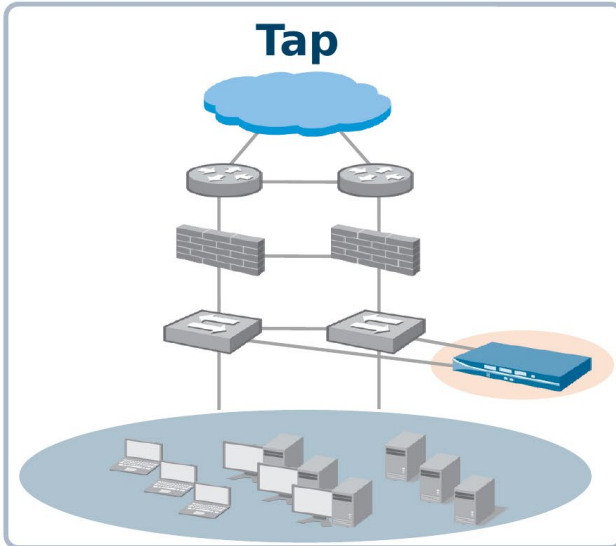
LESSON 3

Tap and Virtual Wire Deployments

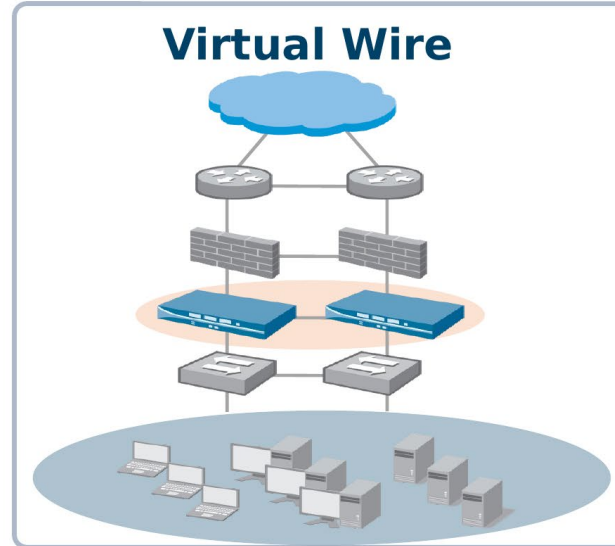


Flexible Deployment Options

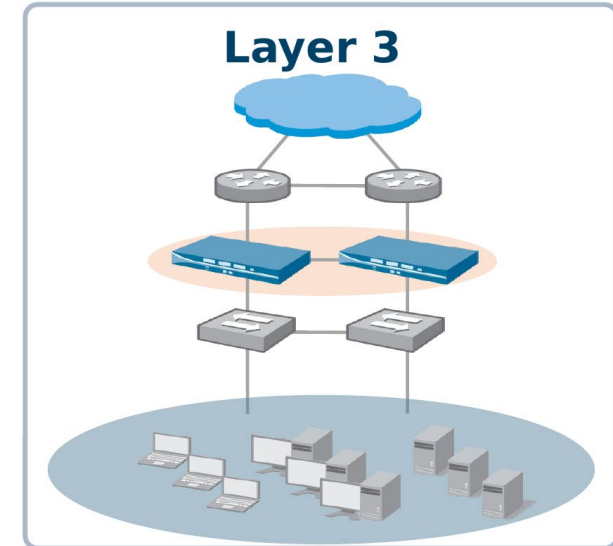
One firewall can mix these deployments on different ports.



Visibility without an inline deployment; used to evaluate and audit existing networks.



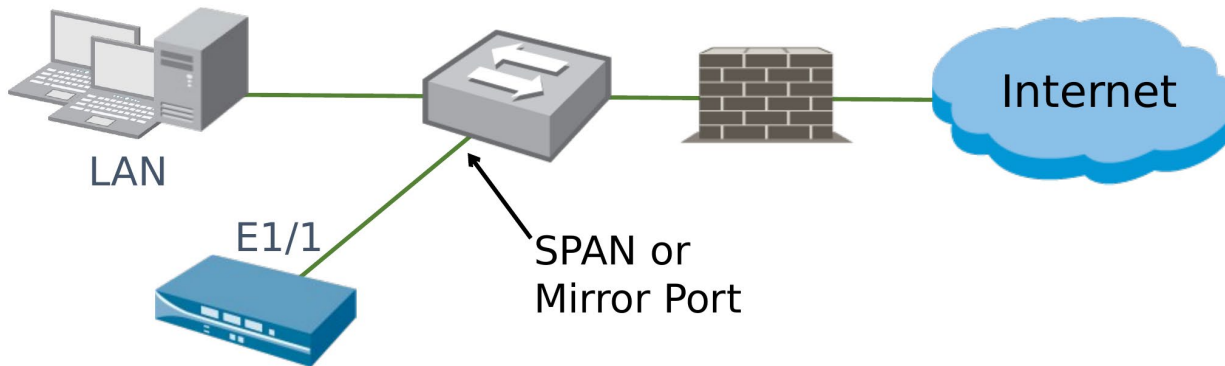
Inline and transparent: App-ID, Content-ID, User-ID, SSL decryption, and NAT.



Everything virtual wire does, plus virtual routers, VPN, and routing protocols.

Tap Interfaces

- Passively monitor a switch's **SPAN or mirror port** — no network changes.
- Traffic reaches the firewall but does not pass through it: **no blocking or shaping**.
- Must be assigned to a **tap zone**.
- Traffic log data informs the Security rules you write later.



Logging still needs policy

Security rules require zones, so a tap interface still needs a tap zone.

Write a rule with the tap zone as both source and destination to log what you see.

Configure a Tap Interface

Network > Interfaces > Ethernet > <interface> — set the type, then choose a tap zone.

Ethernet Interface

Interface Name

ethernet1/3

Comment

Tap interface for monitoring traffic only

Interface Type

Tap

Netflow Profile

None

Config

Advanced

Assign Interface To

Security Zone

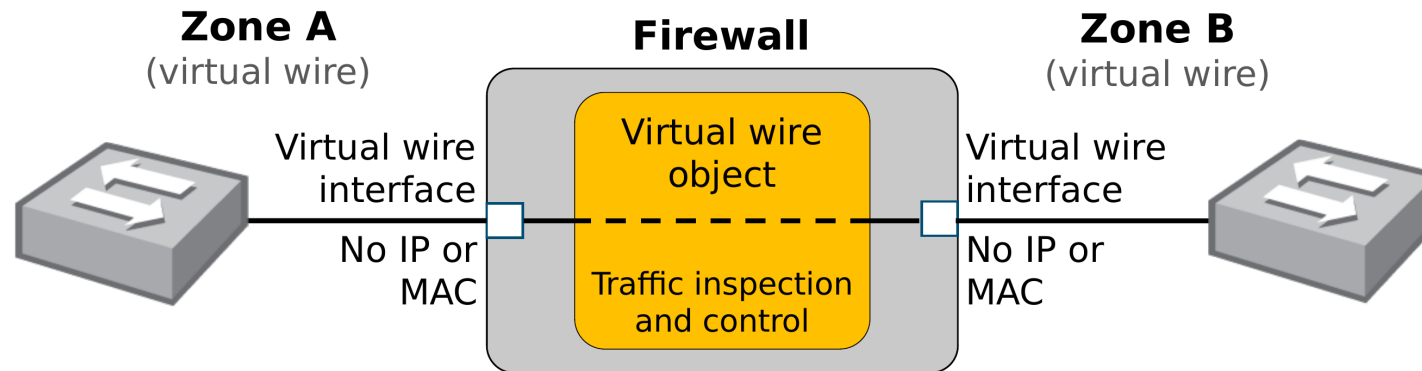
Monitor_Only_Zone

Ethernet VLAN Loopback Tunnel SD-WAN									
INTERFACE	INTERFACE TYPE	MANAGEMENT PROFILE	LINK STATE	IP ADDRESS	VIRTUAL ROUTER	TAG	VLAN / VIRTUAL-WIRE	SECURITY ZONE	SD-WAN INTERFACE PROFILE
ethernet1/1	Layer3	Allow-ping		203.0.113.20/24	VR-1	Untagged	none	Internet	
ethernet1/2	Layer3	Allow-mgt		192.168.1.1/24	VR-1	Untagged	none	Users_Net	
ethernet1/3	Layer3	Allow-mgt		192.168.50.1/24	VR-1	Untagged	none	Extranet	
ethernet1/4				none	none	Untagged	none	none	
ethernet1/5				none	none	Untagged	none	none	
ethernet1/6				none	none	Untagged	none	none	
ethernet1/7				none	none	Untagged	none	none	
ethernet1/8				none	none	Untagged	none	none	
ethernet1/9				none	none	Untagged	none	none	

- **Interface Type:** Tap.
- **Security Zone:** only tap-type zones appear in the list.
- Encrypted SPAN traffic supports only SSL inbound decryption.

Virtual Wire Interfaces

- Binds two interfaces through a **virtual wire object** — a transparent "bump in the wire".
- **No IP or MAC** on either side, so adjacent devices need no changes.
- Full inspection and blocking, including App-ID, Content-ID, User-ID, decryption, and NAT.
- Cannot route, terminate an IPsec VPN, or carry management traffic.



Configure a Virtual Wire

Network > Virtual Wires > Add creates the object; each interface is then set to type Virtual Wire.

Virtual Wire

Name

Vwire_Object

Interface1

ethernet1/4

Interface2

ethernet1/5

Tag Allowed

[0 - 4094]

Enter either integers (e.g. 10) or ranges (100-200) separated by commas. Integer values can be between 0 and 4094.

☐

 Multicast Firewalling

☒

 Link State Pass Through

- **Tag Allowed:** 802.1Q values 0–4094, where 0 means untagged.
- **Link State Pass Through** is on by default; **Multicast Firewalling** is optional.
- Each interface needs a **virtual wire zone** — typically one zone per side.

Ethernet Interface

Interface Name

ethernet1/5

Comment

Vwire for the Danger Zone

Interface Type

Virtual Wire

Netflow Profile

None

Config

Advanced

Assign Interface To

Virtual Wire

Vwire_Object

Security Zone

Danger

Comparing the Deployment Options

	Tap	Virtual Wire	Layer 3
IP address on interface	No	No	Yes
Blocks traffic	No	Yes	Yes
App-ID / Content-ID	Yes	Yes	Yes
NAT	No	Yes	Yes
Routing and VPN	No	No	Yes
Management access	No	No	Yes, with a profile
Network changes needed	None	None	Readdressing likely

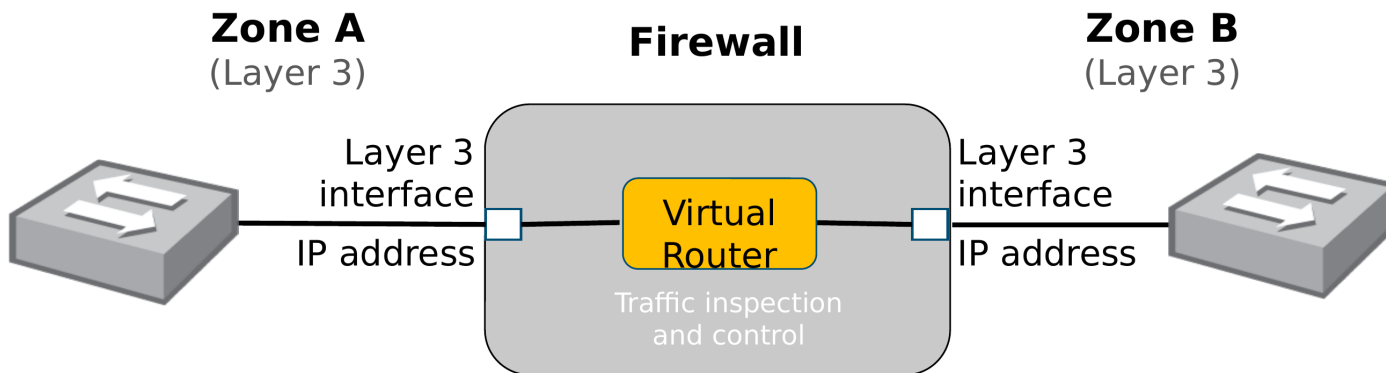
LESSON 4

Layer 3 Interfaces



Layer 3 Interfaces

- Each interface gets an **IP address** and the firewall routes between them.
- Routing requires a **virtual router**.
- Adjacent networks may need readdressing or new gateways.
- Can carry management traffic — because it has an address.



Everything virtual wire does, plus

Virtual routers and dynamic routing protocols.

IPsec VPN termination.

NAT, QoS, decryption, and management access through Interface Management Profiles.

Configure a Layer 3 Interface: Config Tab

Network > Interfaces > Ethernet > <interface> — Config tab.

Ethernet Interface

Interface Name

ethernet1/1

Comment

Interface connected to the Internet

Interface Type

Layer3

Netflow Profile

None

Config

IPv4

IPv6

SD-WAN

Advanced

Assign Interface To

Virtual Router

VR-1

Security Zone

Internet

- **Interface Type:** Layer3.
- **Virtual Router:** interfaces on the same router share one route table.
- **Security Zone:** only Layer 3 zones are listed.
- **Comment:** describe what the port connects to.

Configure a Layer 3 Interface: IPv4 and IPv6

Ethernet Interface ?

Interface Name: ethernet1/1

Comment: Interface connected to the Internet

Interface Type: Layer3

Netflow Profile: None

Config | **IPv4** | IPv6 | SD-WAN | Advanced

☐ Enable SD-WAN

Type: ☒ Static ☐ PPPoE ☐ DHCP Client

☐	IP
☐	203.0.113.20/24

+ Add - Delete ↑ Move Up ↓ Move Down

IP address/netmask. Ex. 192.168.2.254/24

Config | **IPv4** | IPv6 | SD-WAN | Advanced

☐ Enable SD-WAN

Type: ☐ Static ☐ PPPoE ☒ DHCP Client

☒ Enable

☒ Automatically create default route pointing to default gateway provided by server

☐ Send Hostname system-hostname

Default Route Metric: 10

[Show DHCP Client Runtime Info](#)

- **Static:** CIDR notation, e.g. 203.0.113.20/24.
- **DHCP Client:** dynamic address, optional learned default route.
- **PPPoE:** terminate a DSL connection.
- **IPv6:** enable first at `Device > Setup > Session`.

Layer 3 interfaces support dual-stack IPv4 and IPv6.

Configure a Layer 3 Interface: Advanced Tab

The Other Info subtab is where an Interface Management Profile is applied.

The screenshot shows the 'Ethernet Interface' configuration page with the 'Advanced' tab selected. The 'Interface Name' is 'ethernet1/1' and the 'Comment' is 'Interface connected to the Internet'. The 'Interface Type' is 'Layer3' and the 'Netflow Profile' is 'None'. Below these are tabs for 'Config', 'IPv4', 'IPv6', 'SD-WAN', and 'Advanced'. The 'Link Settings' section includes 'Link Speed' (auto), 'Link Duplex' (auto), and 'Link State' (auto). The 'Other Info' section includes 'Management Profile' (Allow-ping), 'MTU' (576 - 1500), and 'Adjust TCP MSS' (unchecked). The 'Adjust TCP MSS' section has 'IPv4 MSS Adjustment' (40) and 'IPv6 MSS Adjustment' (60). At the bottom is an 'Untagged Subinterface' checkbox.

Ethernet Interface

Interface Name: ethernet1/1

Comment: Interface connected to the Internet

Interface Type: Layer3

Netflow Profile: None

Config | IPv4 | IPv6 | SD-WAN | **Advanced**

Link Settings

Link Speed: auto Link Duplex: auto Link State: auto

Other Info | ARP Entries | ND Entries | NDP Proxy | LLDP | DDNS

Management Profile: Allow-ping

MTU: [576 - 1500]

☐ Adjust TCP MSS

IPv4 MSS Adjustment: 40

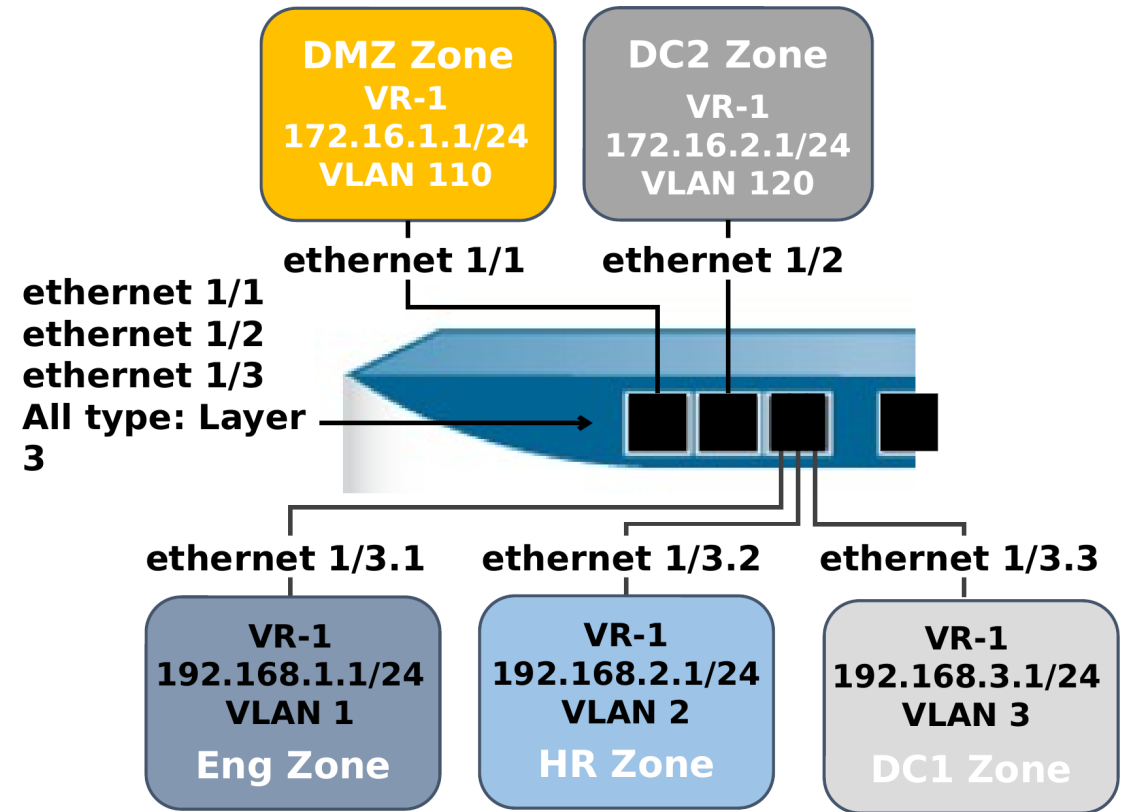
IPv6 MSS Adjustment: 60

☐ Untagged Subinterface

- **Management Profile** — which services the port answers.
- **Link settings**, MTU, and TCP MSS adjustment.
- **ARP / ND entries** — pre-load neighbor caches.
- **NDP Proxy**, **LLDP**, and **DDNS**.

Layer 3 Subinterfaces

- One port, many logical interfaces, each processing traffic by:
- **VLAN tag** (1–4094)
- **VLAN tag plus IP classifier** (source IP)
- **IP classifier** alone, for untagged traffic
- Each subinterface can sit in its **own zone** for more granular rules.



Configure a Layer 3 Subinterface

Network > Interfaces > Ethernet — select the parent and click Add Subinterface.

Layer3 Subinterface

Interface Name

ethernet1/2

.

1

Comment

Layer 3 Subinterface for eth 1/2

Tag

1

Netflow Profile

None

Config

IPv4

IPv6

SD-WAN

Advanced

Assign Interface To

Virtual Router

VR

Security Zone

Users_Net

- **Interface Name**: parent plus a subinterface ID, e.g. `ethernet1/2.1`.
- **Tag**: the 802.1Q VLAN ID. Everything else is configured as a normal Layer 3 interface.

Ethernet | VLAN | Loopback | Tunnel

INTERFACE	INTERFACE TYPE
ethernet1/1	Layer3
ethernet1/2	Layer3
ethernet1/3	Layer3
ethernet1/4	Virtual Wire
ethernet1/5	Virtual Wire
ethernet1/6	
ethernet1/7	
ethernet1/8	
ethernet1/9	

+

 Add Subinterface

-

 Delete

PDF/CSV

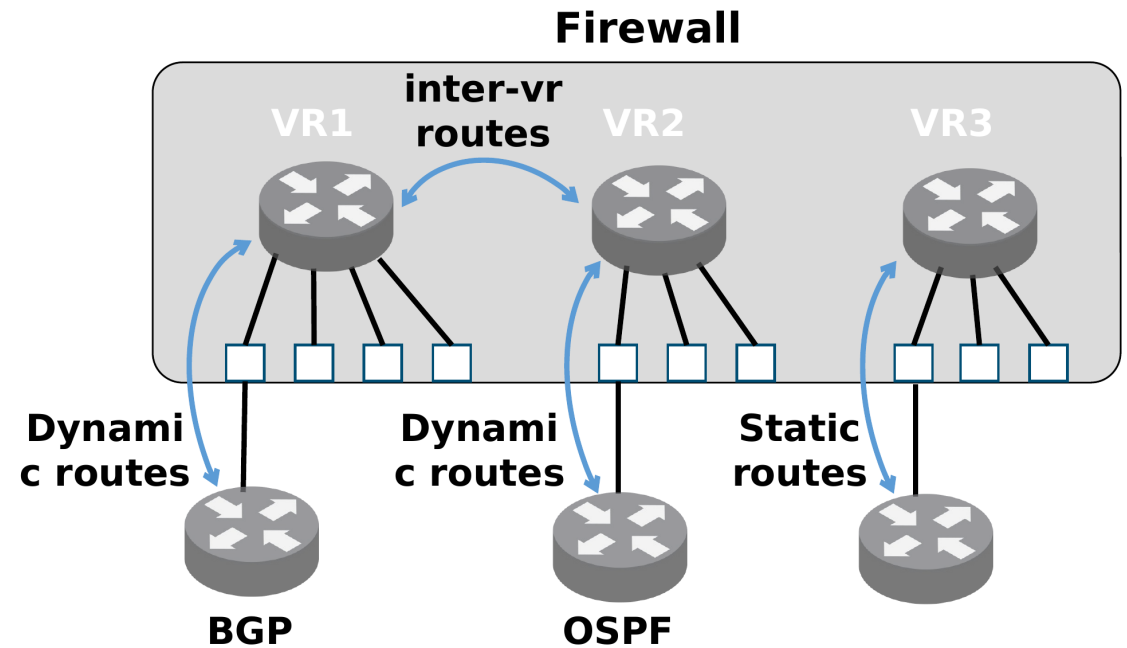
LESSON 5

Virtual Routers and Static Routes



Virtual Routers

- Learns routes: **connected**, **static**, and **dynamic**.
- Dynamic routing: BGPv4, OSPFv2, OSPFv3, RIPv2.
- Multicast: PIM-SM and PIM-SSM.
- **RIB** holds every known route; **FIB** holds the best routes actually used to forward.
- Multiple virtual routers can exchange traffic through **inter-VR routes**.



Virtual Router General Settings

Network > Virtual Routers — name the router and add the interfaces it may forward on.

Virtual Router - VR-1

Router Settings

Static Routes

Redistribution Profile

RIP

OSPF

OSPFv3

BGP

Multicast

Name VR-1

General

ECMP

☐

INTERFACES ^

☐ ethernet1/1

☐ ethernet1/2

☐ ethernet1/3

☒ |

ethernet1/1

ethernet1/2

ethernet1/3

loopback

sdwan

tunnel

vlan

+ Add

- Delete

Administrative Distances

Static

10

Static IPv6

10

OSPF Int

30

OSPF Ext

110

OSPFv3 Int

30

OSPFv3 Ext

110

IBGP

200

EBGP

20

RIP

120



Administrative distance

When two sources offer the same destination, the **lower distance wins**.
Static routes default to 10; OSPF internal to 30; EBGP to 20.

Add a Static Default Route

Network > Virtual Routers > Static Routes > Add — 0.0.0.0/0 is the route of last resort.

Virtual Router - Static Route - IPv4

Name

Firewall-Default-Gateway

Destination

0.0.0.0/0

Interface

ethernet1/1

Next Hop

IP Address

203.0.113.1

Admin Distance

10 - 240

Metric

10

Route Table

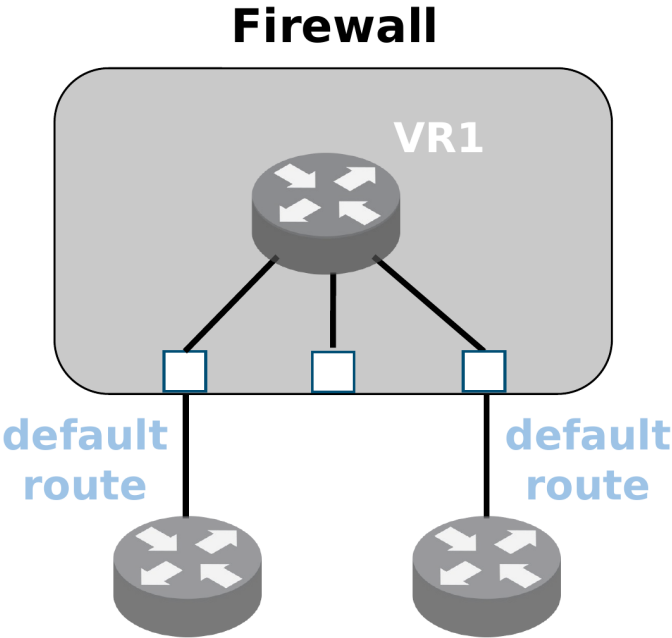
Unicast

BFD Profile

Disable BFD

- **Destination:** 0.0.0.0/0 or ::/0.
- **Interface:** the egress port — and its zone.
- **Next Hop:** IP address, next VR, Discard, or None.
- **Metric** picks between routes from the same source.
- **Route Table:** unicast, multicast, both, or No Install.

Multiple Default Routes and Path Monitoring



Static routes default to administrative distance 10.

☒ Path Monitoring

Failure Condition ☒ Any ☐ All Preemptive Hold Time (min) 2

☐	NAME	ENABLE	SOURCE IP	DESTINATION IP	PING INTERVAL(SEC)	PING COUNT
+ Add - Delete						

Path Monitoring Destination ?

Name Monitor Server IP past Router

☒ Enable

Source IP 203.0.113.20/24

Destination IP 8.8.8.8

Ping Interval(sec) 3

Ping Count 5

- Lowest **metric** wins; the firewall fails over and fails back.
- Pings stable upstream destinations — **Any** or **All** failure condition.
- Failed routes leave the RIB and FIB; a hold timer restores them.

Troubleshoot Routing

Network > Virtual Routers > More Runtime Stats shows what the router actually knows.

☐	NAME	INTERFACES	CONFIGURATION	RIP	OSPF	OSPFV3	BGP	MULTICAST	RUNTIME STATS
☐	default		ECMP status: Disabled						More Runtime Stats
☐	VR-1	ethernet1/1 ethernet1/2 ethernet1/3	Static Routes: 1 ECMP status: Disabled						More Runtime Stats

Virtual Router - VR-1						
Routing RIP OSPF OSPFv3 BGP Multicast BFD Summary Information						
Route Table Forwarding Table Static Route Monitoring						
Route Table ☒ Unicast ☐ Multicast Display Address Family IPv4 and IPv6						
7 items						
DESTINATION	NEXT HOP	METRIC	WEIGHT	FLAGS	AGE	INTERFACE
0.0.0.0/0	203.0.113.1	10		A S		ethernet1/1
192.168.1.0/24	192.168.1.1	0		A C		ethernet1/2
192.168.1.1/32	0.0.0.0	0		A H		
192.168.50.0/24	192.168.50.1	0		A C		ethernet1/3
192.168.50.1/32	0.0.0.0	0		A H		
203.0.113.0/24	203.0.113.20	0		A C		ethernet1/1
203.0.113.20/32	0.0.0.0	0		A H		
Refresh A:active, ?:loose, C:connect, H:host, S:static, ~:internal, R:rip, O:OSPF, B:bgp, Oi:OSPF intra-area, Oo:OSPF inter-area, O1:OSPF ext-ty						

Tab

Shows

Route Table

The RIB — all known routes

Forwarding Table

The FIB — where traffic goes

Static Route Monitoring

Path monitor status

show routing route

LESSON 6

Interface Management Profiles



Interface Management Profile

Network > Network Profiles > Interface Mgmt > Add

Interface Management Profile

Name: HTTPS_SSH_Ping_SNMP_RPs

Administrative Management Services

- ☐ HTTP
- ☒ HTTPS
- ☐ Telnet
- ☒ SSH

Network Services

- ☒ Ping
- ☐ HTTP OCSP
- ☒ SNMP
- ☒ Response Pages
- ☐ User-ID
- ☐ User-ID Syslog Listener-SSL
- ☐ User-ID Syslog Listener-UDP

PERMITTED IP ADDRESSES

+ Add - Delete

Ex. IPv4 192.168.1.1 or 192.168.1.0/24 or IPv6 2001:db8:123:1::1 or 2001:db8:123:1::/64

- With no profile, the interface answers **no management services**.
- **Administrative**: HTTP, HTTPS, Telnet, SSH.
- **Network**: Ping, HTTP OCSP, SNMP, Response Pages, User-ID.
- **Permitted IP Addresses** limit who may connect; blank means anyone.
- Applies to Layer 3, loopback, VLAN, and tunnel interfaces.

The Lab Profiles: Allow-ping and Allow-mgt

Allow-ping → ethernet1/1 (Internet)

- Administrative services: **none**
- Network services: **Ping**
- Lets SecOps confirm the outside interface is up

Allow-mgt → ethernet1/2 and 1/3

- Administrative: **HTTPS, SSH**
- Network: Ping, SNMP, Response Pages, User-ID
- Encrypted protocols only — HTTP and Telnet stay off

Read the warning: attaching administrative services exposes the management interface to anyone who can reach that port. Never do it on an internet-facing interface.

Real-World Anchor: CVE-2024-0012

- An authentication bypass in the PAN-OS **management web interface**, exploited in the wild in late 2024.
- Highest risk: firewalls whose management interface was **reachable from the internet**.
- Key mitigation: restrict management access to **trusted internal addresses**.
- On a data-plane port, that decision lives in the **Interface Management Profile**.



SECURITY FOCUS

Never put HTTPS or SSH on an outside interface, and use Permitted IP Addresses to limit internal management to an admin subnet.

Why it matters: One exposed management page turned a web bug into full control of the firewall.

Verify Connectivity and Access

From the firewall CLI, always ping from a data-plane address:

```
admin@firewall-a> ping source 192.168.1.1 host 192.168.1.20
admin@firewall-a> ping source 192.168.50.1 host 192.168.50.150
admin@firewall-a> ping source 203.0.113.20 host 8.8.8.8
```

- Without `source`, the firewall pings from its MGT address.
- From Client-A, `ping 192.168.1.1` and `ssh admin@192.168.1.1` fail before the profile — and succeed after.
- Nothing takes effect until you **commit**.

Segmentation Against Ransomware

Most ransomware follows the same arc — zones decide how far it gets after the first host.

Stage	What the attacker does	What helps
Initial access	Phishing, stolen credentials, unpatched VPNs	Patching, MFA, no exposed management
Lateral movement	Hops toward servers, domain controllers, backups	Zones with default interzone deny
Data theft	Copies data out before encrypting	Inspect and restrict outbound traffic
Encryption	Encrypts everything it can reach	Backups isolated in their own zone

For your memo: if the backup server is reachable from the users' zone, it is reachable by the ransomware too.

Demo Inventory

In order, on the NETLAB+ pod:

```
admin@firewall-a> show interface all
admin@firewall-a> show routing route
admin@firewall-a> ping source 192.168.1.1 host 192.168.1.20
admin@firewall-a> ping source 203.0.113.20 host 8.8.8.8
lab-user@client-a:~$ ping 192.168.1.1
lab-user@client-a:~$ ssh admin@192.168.1.1
```

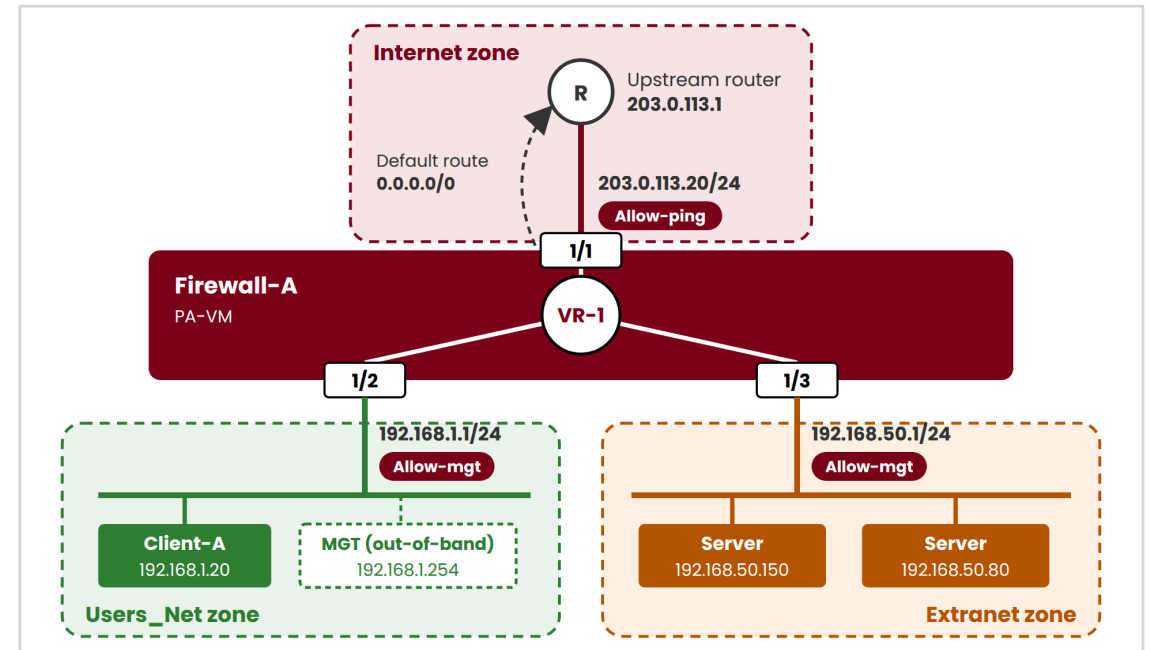


- Show the **Zones** page and the interface list's **Management Profile** column.
- Repeat the Client-A tests after applying **Allow-mgt** and committing.

Hands-On Lab 4: Security Zones

NETLAB+ · Palo Alto Networks Firewall 11.0 Essentials · Lab 4

- Load and commit `edu-210-11.0a-04.xml`.
- Build three Layer 3 interfaces and rename the virtual router **VR-1**.
- Add the **Firewall Default Gateway** route to 203.0.113.1.
- Create **Internet**, **Users_Net**, and **Extranet** zones; commit.
- Test each zone, then apply **Allow-ping** and **Allow-mgt** and retest.



Assignment 1: The State of Ransomware Memo

Choose one case study from the State of Ransomware document in Blackboard.

- **Incident summary:** who was hit, how attackers got in, how they spread, and the impact.
- **Root causes:** exposed services, weak credentials, a flat network, reachable backups.
- **Prioritized recommendations,** including at least one using zones, interzone deny, or management profiles.
- **Recovery readiness** and **sources,** such as the CISA #StopRansomware Guide.



Always check Blackboard for the due date and rubric.

Review Questions

1. Which two are supported routing protocols on a virtual router? OSPF · IGRP · EGP · BGP
2. Which three interface types are valid? FC · Layer3 · FCoE · Tap · Virtual wire
3. Which two interface types can join a Layer3 zone? Tunnel · Virtual wire · Tap · Loopback
4. Which interface type enables passive monitoring? Virtual wire · Tap · Loopback · Tunnel
5. True or false: a Layer 3 interface can be dual-stack IPv4 and IPv6.



Key Takeaways



Source and destination zones are decided at session setup, before any rule is checked.



Traffic moves freely inside a zone; nothing crosses zones until a rule allows it.



Tap sees, virtual wire blocks without addresses, Layer 3 routes.



A Layer 3 interface needs a type, an IP, a virtual router, and a zone.



The RIB is everything the router knows; the FIB is what it actually uses.



Data-plane interfaces answer nothing until a management profile says so — keep it off the outside.

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Questions?

Unit 5 · Connecting the Firewall with Security Zones